IDENTIDADES TRIGONOMÉTRICAS

— YisusReveck —

Recíprocas

$$\text{sen } \alpha = \frac{1}{\csc \alpha} \qquad \csc \alpha = \frac{1}{\text{sen } \alpha}$$

$$\cos \alpha = \frac{1}{\sec \alpha} \qquad \sec \alpha = \frac{1}{\cos \alpha}$$

$$\tan \alpha = \frac{1}{\cot \alpha} \qquad \cot \alpha = \frac{1}{\tan \alpha}$$

Pitagóricas

$$\text{sen}^2\alpha + \cos^2\alpha = 1$$

$$1 + \tan^2\alpha = \sec^2\alpha$$

$$1 + \cot^2\alpha = \csc^2\alpha$$

Cociente

$$\tan \alpha = \frac{\text{sen } \alpha}{\cos \alpha} \qquad \cot \alpha = \frac{\cos \alpha}{\text{sen } \alpha}$$

Ángulos opuestos

$$\text{sen}(-\alpha) = -\text{sen } \alpha$$

$$\cos(-\alpha) = \cos \alpha$$

$$\tan(-\alpha) = -\tan \alpha$$

Ángulos dobles

$$\text{sen}(2\alpha) = 2\,\text{sen } \alpha \cos \alpha$$

$$\cos(2\alpha) = \cos^2\alpha - \text{sen}^2\alpha$$

$$\tan(2\alpha) = \frac{2\tan \alpha}{1 - \tan^2\alpha}$$

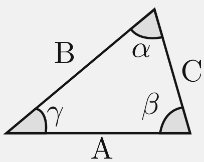
Suma y resta

$$\text{sen}(\alpha \pm \beta) = \text{sen } \alpha \cos \beta \pm \cos \alpha \text{ sen } \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \text{sen } \alpha \text{ sen } \beta$$

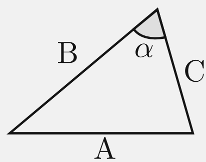
$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$$

Ley de senos



$$\frac{A}{\text{sen } \alpha} = \frac{B}{\text{sen } \beta} = \frac{C}{\text{sen } \gamma}$$

Ley de cosenos



$$A^2 = B^2 + C^2 - 2\ BC \cos \alpha$$